NATIONAL UNIVERSITY OF SINGAPORE
Department of Statistics and Data Science

ST5202: Applied Regression Analysis

TUTORIAL 7 (Solution)

1. For the **SMSA data set** considered in Tutorial 3.

   (i) Suppose that people are concerned with the differences in serious crimes (y) among the regions z and think that the following variables might affect the rate of serious crimes: population density (x_1, total population divided by land area), total personal income (x_2) and percent high school graduates (x_3).

   (ia) Suggest an appropriate regression model for dealing with the concern.

   Here the prime purpose of the study is to compare the serious crimes in different regions, the variables x_1, x_2, x_3 are concomitant variables. The appropriate model is as follows:

$$y = \beta_0 + \sum_{j=1}^{3} \alpha_j x_j + \boxed{\sum_{k=2}^{4} \beta_k z_k} + \epsilon,$$

$\beta_2 \underset{0/1}{z_2} + \beta_3 z_3 + \beta_4 z_4$

   where z_k's are the dummy variable for regions.

   (ib) A model is fitted to the data, which yields the following results:

```
Estimated coefficients:

 (Intercept)          x1          x2          x3          z2          z3          z4
-7747.027762  2443.184486    8.605784  -131.693527  9142.365366 17408.660144 26821.270021

Estimated variance matrix:

(Intercept)            x1        x2            x3            z2            z3            z4
(Intercept)  1.662488e+08 -5156333.6561 169.1947 -2895352.2202 -2064060.3821 -2.026821e+07 17186007.9838
x1          -5.156334e+06  2008009.2788 -99.3373    62356.9546  1384486.9157  1.852863e+06  1139626.9088
x2           1.691947e+02      -99.3373   0.0265       -5.0192      -24.1182 -7.380800e+00       -5.2891
x3          -2.895352e+06    62356.9546  -5.0192    55077.8296  -198314.2332  1.378800e+05  -566625.7660
z2          -2.064060e+06  1384486.9157 -24.1182  -198314.2332 20924969.5983  1.167900e+07 14189529.1919
z3          -2.026821e+07  1852862.9274  -7.3808   137880.0487 11678998.8811  1.843041e+07 10881673.2113
z4           1.718601e+07  1139626.9088  -5.2891  -566625.7660 14189529.1919  1.088167e+07 28370530.6398


Analysis of Variance Table
       Df     Sum Sq    Mean Sq   F value    Pr(>F)
x1          1 1.9396e+11 1.9396e+11  676.3277 < 2.2e-16
x2          1 8.3193e+11 8.3193e+11 2900.8761 < 2.2e-16
x3          1 2.6892e+08 2.6892e+08    0.9377    0.3346
z           3 9.7021e+09 3.2340e+09   11.2768 1.208e-06
Residuals 134 3.8429e+10 2.8679e+08
```

   Give the model which was fitted.

   The model is as given in (ia).

$y = \beta_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \beta_2 z_2 + \beta_3 z_3 + \beta_4 z_4$

(ic) From the results given in (ib), test whether there is any difference in serious crimes among the regions: give the p-value of the test and draw your conclusion.

The question amounts to test

$$H_0 : \boldsymbol{\beta} = (\beta_2, \beta_3, \beta_4)^\top = 0, \ v.s. \ H_1 : \boldsymbol{\beta} \neq 0.$$

$\begin{pmatrix} \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ or not

It can be tested by the F-value corresponding to z in the ANOVA table which is 11.2768 with a p-value `1.208e-06`.
From the small p-value, we can reject the null hypothesis and conclude that the difference among the regions are significant.

(id) Test at level $\alpha = 0.05$ whether or not there is significant difference (1) between region 2 and 4, (2) between region 1 and 3, and (3) between region 1 and 4.

$\beta_2 - \beta_4 = 0$ or not

For (1), it amounts to test

$\beta_3 = 0$ or not

$\beta_4 = 0$ or not

$$H_0 : \beta_2 - \beta_4 = 0, \ H_1 : \beta_2 - \beta_4 \neq 0.$$

From the estimated coefficients and the estimated variance matrix, the t-statistic is computed as

$$T_1 = \frac{9142.3654 - 26821.2700}{\sqrt{20924969.5983 + 28370530.6398 - 2 * 14189529.1919}} = -3.8656.$$

The p-value is computed as `2*pt(3.8656,134,lower.tail=FALSE)` = `0.000172`.

For (2) and (3), it is equivalent ot test:

$$H_{0j} : \beta_j = 0, \ v.s. \ H_{1j} : \beta_j \neq 0, j = 3, 4.$$

The t-statistics are computed as

$$\begin{aligned} T_2 &= 17408.6601/\sqrt{18430410} = 4.055, \\ T_3 &= 26821.270/\sqrt{28370530} = 5.036. \end{aligned}$$

The p-values are computed as `2*pt(abs(T2),134,lower.tail=FALSE) = 8.46e-05`, `2*pt(abs(T2),134,lower.tail=FALSE) = 1.51e-06`.
When the overall type I error rate is to be controlled, the individual p-values must be multiplied by 3. The Bonforroni p-values of the three tests are less than 0.05, all the null hypotheses are rejected.

(ii) Suppose that people want to find out how the three variables, x_1, x_2 and x_3, affect the serious crimes and how their effects differ in different regions.

$x_1 z_2$
$x_1 \leftarrow x_1 z_3$
$x_1 z_4$
x_2
x_3

(iia) What is the appropriate model for this purpose?

The model should be able to compare both the intercepts and slopes of the regression function within the four regions. A model including all the main-effect terms as well as the interaction between the x's and z is appropriate. The model is as follows:

$$y = \beta_0 + \sum_{j=1}^{3} \alpha_j x_j + \sum_{k=2}^{4} \beta_k z_k + \sum_{k=2}^{4}\sum_{j=1}^{3} \gamma_{kj} z_k x_j + \epsilon.$$

The regression function expressed separately for each of the regions are:

$$y = \begin{cases} \beta_0 + \sum_{j=1}^{3} \alpha_j x_j + \epsilon, & \textit{region 1}; \\ (\beta_0 + \beta_2) + \sum_{j=1}^{3} (\alpha_j + \gamma_{2j}) x_j + \epsilon, & \textit{region 2}; \\ (\beta_0 + \beta_3) + \sum_{j=1}^{3} (\alpha_j + \gamma_{3j}) x_j + \epsilon, & \textit{region 3}; \\ (\beta_0 + \beta_4) + \sum_{j=1}^{3} (\alpha_j + \gamma_{4j}) x_j + \epsilon, & \textit{region 4}. \end{cases}$$

(iib) People believe that the population density (x_1) and total personal income (x_2) have the same effects among different regions and that they would like to find out specifically whether the effect of percent high school graduates (x_3) is different among the regions. State an appropriate model for this purpose. Fit the model and test the significance of the difference: (1) state the hypotheses in terms of the parameters in the model, (2) provide a method for testing the hypotheses and draw your conclusion.

The appropriate model is

$134 - 3 = 131$

$$y = \beta_0 + \sum_{j=1}^{3} \alpha_j x_j + \sum_{k=2}^{4} \beta_k z_k + \sum_{k=2}^{4} \gamma_{k3} z_k x_3 + \epsilon.$$

(1) Let $\boldsymbol{\gamma} = (\gamma_{23}, \gamma_{33}, \gamma_{43})^\top$. *The hypotheses are:*

$$H_0 : \boldsymbol{\gamma} = 0, \;\; H_1 \boldsymbol{\gamma} \neq 0.$$

$\begin{pmatrix} \gamma_{23} \\ \gamma_{33} \\ \gamma_{43} \end{pmatrix} = 0$ or not

(2) To test the above hypotheses, the test statistic is given by

$$F = \hat{\boldsymbol{\gamma}}^\top \hat{\Sigma}_\gamma \boldsymbol{\gamma} / 3,$$

which follows a F-distribution with d.f. 3 and 131. $= 134 - 3$ (new cross term)

The R codes for obtaining $\hat{\boldsymbol{\gamma}}$, $\hat{\Sigma}_\gamma$, F *and the p-value are as follows:*

```
p.int = lm(y~x1+x2+x3*z); summary(p.int); b=p.int$coef; V=vcov(p.int)
gm = b[8:10]; v.gm= V[8:10, 8:10]; gm; v.gm
F= t(gm)%*%solve(v.gm)%*%gm/3
p.value = pf(F,3,131,lower.tail=FALSE)
```

The computed results are:

```
Estimated gamma's:

    x3:z2     x3:z3     x3:z4
584.97348 117.00861 -98.95316

Estimated variance matrix:

         x3:z2    x3:z3    x3:z4
x3:z2 615301.5 353163.2 361453.9
x3:z3 353163.2 482326.4 362571.2
x3:z4 361453.9 362571.2 666665.6

F-value:
 0.3358275

P-value:
 0.7994507
```

With the big p-value, the null hypothesis is not rejected. It is concluded that there is no evidence to show that x_3 has different effects across the regions.

2. In a Study on the Efficacy of Nosocomial Infection Control (SENIC), it is to determine whether infection surveillance and control programs have reduced the rate of nosocomial (hospital acquired) infection in the United States hospitals. The SENIC data provided in the file: `SENIC.txt`, on Canvas, consists of a random sample of 113 hospitals. The variables of the data set are as follows:

```
y: Infection risk;                x1:Average days of patients stayed;
x2:Average age of patients;       x3:Routine culturing ratio;
x4:Routine chest X-ray ratio;     x5:Number of beds;
x6:Medical school affiliation;    x7:Region (1,2,3,4);
x8:Average daily census;          x9:Number of nurses;
x10:Available facilities and services.
```

(i) Consider the relationship between infection risk (y, the response variable) and the

following three predictor variables: patient's average age (x_2), average daily census (x_8) and available facilities and services (x_{10}).

(ia) For each of the predictor variables, conduct an appropriate test at level $\alpha = 0.05$ to find out whether or not the variable has the same effect in the four regions.

An appropriate model for dealing with this problem is the model including all main effect terms of the x's and the region (R) as well as interaction terms between the x's and R. Whether or not a x-variable has the same effect in the four regions is equivalent to the regression parameter corresponding to the interaction terms between that x and R. The test is the F-test. The F-statistic is computed though the Wald statistic approach. The R codes for the computation are as follows:

```
q2.d = read.table("D://Rsession/SENIC.txt",header =TRUE)

y = q2.d$y; z1=q2.d$x2; z2=q2.d$x8; z3=q2.d$x10; R=factor(q2.d$x7)

q2.fit=lm(y~z1+z2+z3+R+z1:R+z2:R+z3:R)

b=q2.fit$coef; b1=b[8:10]; b2=b[11:13]; b3=b[14:16]
V=vcov(q2.fit);V1=V[8:10,8:10]; V2=V[11:13,11:13]; V3=V[14:16,14:16]
F1=t(b1)%*%solve(V1)%*%b1/3; F2=t(b2)%*%solve(V2)%*%b2/3;
F3=t(b3)%*%solve(V3)%*%b3/3;
```

The F-values corresponding to x_2, x_8 and x_{10} are respectively 1.4789, 0.7530 and 0.1092. The p-values are: 0.2251, 0.5233 and 0.9546, respectively. With these p-values, there is no significant evidence to show that these variables have different effect across the regions.

(ib) Test at level $\alpha = 0.05$ whether or not there are any differences among the four regions after adjusting for the effect of the three predictor variables.

$z_1 + z_2 + z_3 + R$

factor

To deal with this problem, the appropriate model is the one with only the main-effect terms. The anova table from fitting the model is given below:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
z1	1	0.000	0.0002	0.0002	0.98982	
z2	1	29.393	29.3930	19.9924	1.958e-05	***
z3	1	6.842	6.8422	4.6539	0.03324	*
R	3	9.302	3.1008	2.1091	0.10343	
Residuals	106	155.842	1.4702			

The region effect is tested by the F-value corresponding R in the above table. Since the p-value 0.10343 is larger than 0.05, there are no significant differences among the four regions.

The F-statistic can also be computed though the Wald statistic approach by using the R codes below:

```
m.f = lm(y~z1+z2+z3+R)
```

```
b=m.f$coef[5:7]; V=vcov(m.f)[5:7,5:7]
F=t(b)%*%solve(V)%*%b/3;
```

(ii) Fit a regression model with response variable y, the infection risk, and predictor variables: patient's average age (x_2), average daily census (x_8), available facilities and services (x_{10}), including all the main-effect terms and all the two-variable interaction terms.

$y \sim z_1 + z_2 + z_3 + z_1 z_2 + z_2 z_3 + z_1 z_3$

The model is fitted which yields the following results:

```
i.f = lm(y~z1*z2+z1*z3+z2*z3); summary(i.f);
```

```
            Df  Sum Sq Mean Sq F value    Pr(>F)
z1           1   0.000  0.0002  0.0002  0.989136
z2           1  29.393 29.3930 22.7522 5.912e-06 ***
z3           1   6.842  6.8422  5.2963  0.023326 *
z1:z2        1   9.331  9.3309  7.2228  0.008360 **
z1:z3        1   6.810  6.8098  5.2712  0.023648 *
z2:z3        1  12.065 12.0652  9.3393  0.002838 **
Residuals  106 136.939  1.2919
```

$\begin{pmatrix}\beta_{12}\\ \beta_{23}\\ \beta_{13}\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$

(iia) Use a F-test at level $\alpha = 0.05$ to test whether or not there are any interaction effects.

The F-statistic for the test is computed by

```
b=i.f$coef[5:7]; V=vcov(i.f)[5:7,5:7]
F=t(b)%*%solve(V)%*%b/3; p.value = pf(F,3,106,lower.tail=FALSE)
```

which yields: $F = 7.2778$, $p = 0.0001743$. *At level 0.05, the interaction effects are significant.*

(iib) If the F-test in (iia) is significant, which interaction effects are significant at level $\alpha = 0.05$, controlling the overall type I error rate by Bonferroni criteria.

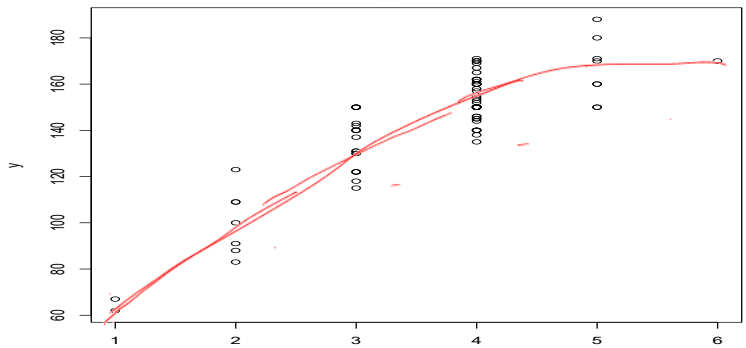
Multiply the p-values for the interaction terms in the summary table by 3, the Bonferroni p-values are obtained as

$$0.02508, 0.0709, 0.0085.$$

At level 0.05, the interaction between x_2 (z_1) *and* x_8 (z_2), *and between* x_8 (z_2) *and* x_{10} (z_3) *are significant, but that between* x_2 *and* x_{10} *is not.*

3. In 1981, $n = 78$ bluegills were randomly sampled from Lake Mary in Minnesota. The researchers measured and recorded the following data: length of the fish (y, in mm) and age of the fish (x, in years). The data is given in `bluegills.txt` on Canvas.

(i) Draw a scatter plot of y against x. Suggest a proper regression model for the data.



The plot is given above. The plot shows a quadratic curvature. A quadratic regression model can be suggested.

(ii) Fit the model and test whether or not there is a significant relation between y and x at level $\alpha = 0.05$. Is R^2 of the model the same as the square of the Pearson's correlation coefficient between y and x? Make comment.

The fitted results are given below:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  147.608      1.472 100.270  < 2e-16 ***
x.c           19.809      1.431  13.845  < 2e-16 ***
I(x.c^2)      -4.719      0.944  -4.999 3.67e-06 ***
Residual standard error: 10.91 on 75 degrees of freedom
Multiple R-squared:  0.8011,    Adjusted R-squared:  0.7958
F-statistic: 151.1 on 2 and 75 DF,  p-value: < 2.2e-16
```

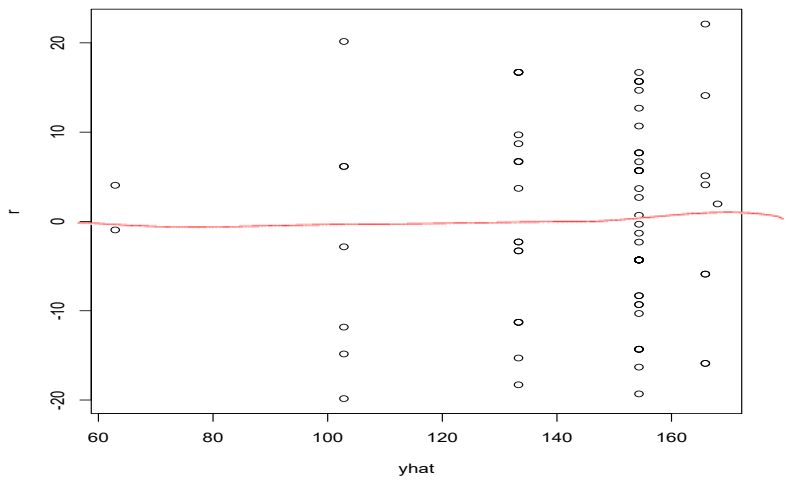
$y \sim x$

$y \sim x + x^2$

The significant F-test has a p-value less than `2.2e-16`*. The regression relationship is significant, i.e., there is a significant relation between y and x.*

The R^2 is not the same as the square of the Pearson's correlation coefficient between y and x. The R^2 of a regression model measures the correlation of the response with all covariates in the model. In the current case, the covariates in the model includes x and x^2. The R^2 equals the square of the multiple correlation coefficient between y and (x, x^2).

(iii) Plot the residuals against the fitted values and comment on the appropriateness of the model.



The plot is given above. No patterns or trends show up in the plot, which indicates the appropriateness of the model.